

Pilot Test Report: Welfare Chatbot Prototype

Sample Size: 12 Users

1. Objective

To validate the usability, comprehension, and technical stability of the AI-powered Multilingual Welfare Scheme Chatbot over a simulated low-bandwidth WhatsApp connection.

2. Methodology

12 participants were selected representing three primary personas:

- Group A (Farmers):** 4 participants (Mixed Hindi/Local dialect).
- Group B (Gig Workers/Daily Wage):** 4 participants (Hinglish/Basic English).
- Group C (Women Head of Households):** 4 participants (Primary Hindi).

Task: Participants were asked to message the Twilio WhatsApp Sandbox number and find out what government schemes they might be eligible for.

3. Success Metrics & Results

Metric	Target	Result	Observation
1. Eligibility Flow Completion Rate	\$\ge\$ 80% completion without dropping off	91% (11 of 12 users completed)	Only one user dropped off at the 3rd question due to confusion over the term "Socio-Economic Caste Census," prompting us to simplify the LLM's system prompt to explain BPL (Below Poverty Line) in simpler terms instead.
2. Comprehension Score	\$\ge\$ 70% of users can name at least one required document	83% (10 of 12 users)	The bulleted checklist generated by the LLM at the end of the flow was highly effective. Users appreciated that the bot explicitly told them to gather their Aadhaar Card and Bank Passbook .
3. Response Time & Technical Stability	< 3 seconds median response time on 2G/3G simulation	2.4 seconds (median)	The FastAPI webhook processed requests instantly. The slight delay was entirely bounded by the Gemini API inference time, which remained well within

Metric	Target	Result	Observation
			acceptable conversational limits.

4. Qualitative Feedback & Edge Cases Handled

Code-Mixing Success

User Input: *"mera ration card kho gaya hai aur mere paas zameen nahi hai"* (My ration card is lost and I have no land).

Bot Handling: The LLM successfully parsed the negative constraint (no land $\rightarrow$ eliminated PM-KISAN) and the missing document (lost ration card). It suggested applying for Ayushman Bharat via Aadhaar instead and advised on re-applying for a Ration Card.

Hallucination Prevention (RAG validation)

User Input: *"Mujhe free laptop kab milega sarkar se?"* (When will I get a free laptop from the government?)

Bot Handling: The system prompt successfully restrained the AI. The bot replied: *"Maaf kijiye, mere paas abhi sirf in 8 schemes ki jaankari hai (PM-KISAN, Ayushman Bharat...)"* It did not hallucinate a fake laptop scheme.

5. Next Steps for Production

- Voice Integration:** Add Twilio's WhatsApp Voice note transcription so users who cannot type can send voice notes.
- State-level Expansion:** Add state-specific schemes (e.g., Rythu Bandhu for Telangana, Kalia for Odisha) based on user location pings.
- Local API Integration:** Transition from Gemini to an open Indic model like AI4Bharat IndicTrans2/Sarvam-1 deployed on edge for offline fallback capabilities.